

Sana'a University

Faculty of Computer and Information Technology

Department of Cyber Security

RESEARCH PROPOSAL

A Hybrid Methodology for Integrating Real-Time Attack Surface Management with Network Traffic Analysis to Generate AI-Enhanced Detection Rules for Reducing Attack Detection Time in Security Operations Centers

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Abstract

Security Operations Centers (SOCs) play a critical role in monitoring, detecting, analyzing, and responding to cybersecurity threats. However, modern SOC environments still face several challenges, including delayed attack detection, high false positive rates, alert fatigue, and weak correlation between asset exposure information and real-time network behavior. Traditional detection approaches often rely on static signatures, generic correlation rules, or isolated alerts, which may not provide sufficient context for detecting multi-stage or evolving attacks.

This research proposes a hybrid methodology that integrates real-time Attack Surface Management (ASM) data with network traffic analysis to generate AI-enhanced dynamic detection rules. The proposed methodology combines information about exposed assets, open ports, running services, vulnerabilities, and asset criticality with network telemetry such as source and destination IP addresses, ports, protocols, connection frequency, packet counts, byte counts, and time-window behavior. By correlating these data sources, the methodology aims to identify high-risk scenarios where suspicious traffic targets exposed, vulnerable, or critical assets.

Artificial intelligence is used as a supporting component for risk scoring, prioritization, and rule generation. The study does not aim to replace SOC analysts or existing SIEM and IDS/IPS platforms. Instead, it aims to provide a structured and interpretable approach that helps analysts generate more contextual detection rules. The proposed methodology will be evaluated by comparing it with traditional static detection rules using metrics such as Mean Time to Detect (MTTD), false positive rate, precision, recall, F1-score, detection rate, and rule interpretability.

The expected contribution of this research is a SOC-oriented methodology that improves detection context, reduces detection delay, supports false positive reduction, and generates dynamic detection rules based on both attack surface visibility and observed network behavior.

Keywords

Security Operations Center; Attack Surface Management; Network Traffic Analysis; SIEM; IDS/IPS; Artificial Intelligence; Dynamic Detection Rules; Mean Time to Detect; False Positive Reduction; Cybersecurity.

Executive Research Snapshot

Item	Description
Proposed Title	A Hybrid Methodology for Integrating Real-Time Attack Surface Mapping with Network Traffic Analysis to Generate AI-Enhanced Detection Rules for Reducing Attack Detection Time in SOC Centers.
Research Area	Cybersecurity, Security Operations Centers, Attack Surface Management, Network Traffic Analysis, Intrusion Detection, AI-Enhanced Detection Engineering.
Research Type	Quantitative Computing Research.
Study Family	Algorithm / System Evaluation.
Core Problem	Current SOC detection approaches often depend on static rules, isolated alerts, and limited correlation between asset exposure and real-time network behavior, which may increase detection delay and false positives.
Main Research Gap	Lack of a hybrid SOC-oriented methodology that integrates real-time ASM data with network traffic analysis and uses artificial intelligence to generate interpretable and dynamic detection rules.
Proposed Solution	A hybrid methodology that combines ASM data, network traffic features, correlation logic, and AI-enhanced risk scoring to generate contextual detection rules for SOC analysts.
Input Data	ASM data such as assets, ports, services, exposure status, vulnerabilities, and asset criticality; network traffic data such as source IP, destination IP, ports, protocols, connection counts, packet counts, byte counts, and time windows.
Main Techniques	Data preprocessing, feature extraction, ASM-to-traffic correlation, AI-enhanced risk scoring, dynamic rule generation, baseline comparison, and quantitative evaluation.

Expected Evaluation Protocol	Comparison between traditional static detection rules and the proposed hybrid detection methodology using controlled datasets, simulated scenarios, or laboratory network data.
Expected Metrics	Mean Time to Detect, Detection Rate, False Positive Rate, Precision, Recall, F1-score, and Rule Interpretability.
Expected Contribution	A structured SOC-oriented methodology that improves detection context, reduces detection time, supports false positive reduction, and produces interpretable AI-enhanced detection rules.
Practical Value	The proposed methodology can support SOC analysts by prioritizing suspicious activities targeting exposed, vulnerable, or critical assets and by helping generate more relevant detection rules.
Scope	The study focuses on defensive detection rule generation using authorized ASM and network traffic data. It does not include offensive exploitation, unauthorized scanning, payload decryption, or full production SOC deployment.

Alignment with Supervisor's Chapter Guides

This proposal is structured according to a quantitative computing research design. The selected topic belongs to the algorithm/system evaluation family because it focuses on designing and evaluating a hybrid SOC-oriented methodology using ASM data, network traffic records, AI-enhanced risk scoring, dynamic detection rule generation, and measurable evaluation metrics.

Supervisor's Guide Requirement	**Alignment in This Proposal**
Problem, Purpose, and Research Questions	Addressed in Chapter 1 through the research background, motivation, research problem, aim, objectives, research questions, and hypotheses.
Literature Review and Research Gap	Addressed in Chapter 2 through a focused review of SOC detection, ASM, network traffic analysis, IDS/IPS and SIEM rules, AI-based detection, and dynamic rule generation.
Quantitative Research Design	Addressed in Chapter 3 by adopting an algorithm/system evaluation design based on measurable detection performance.
Data Source Selection	Addressed in Chapter 3 through the use of ASM data and network traffic data as the main input sources.
Unit of Analysis and Observation	Addressed in Chapter 3 through correlated security scenarios that combine asset exposure and observed network traffic behavior.
Experimental Environment	Addressed in Chapter 3 through the documentation of tools, datasets, preprocessing steps, programming environment, and evaluation scripts.
Experimental Protocol	Addressed in Chapter 3 through baseline comparison, data preprocessing, ASM-to-traffic correlation, AI-enhanced risk scoring, and rule generation.

Evaluation Metrics	Addressed in Chapter 3 through Mean Time to Detect, detection rate, false positive rate, precision, recall, F1-score, and rule interpretability.
Validity and Reproducibility	Addressed in Chapter 3 through threats to validity, assumptions, dataset documentation, and experimental environment documentation.
Ethical Considerations	Addressed in Chapter 3 by avoiding unauthorized scanning, exploitation, or collection of private network data.
Expected Contribution	Addressed across Chapters 1 and 3 through the proposed hybrid methodology for contextual and dynamic SOC detection rule generation.

List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ASM	Attack Surface Management
ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge
C2	Command and Control
CSV	Comma-Separated Values
F1-score	Harmonic Mean of Precision and Recall
FPR	False Positive Rate
IDS	Intrusion Detection System
IP	Internet Protocol
IPS	Intrusion Prevention System
ML	Machine Learning
MTTD	Mean Time to Detect
NTA	Network Traffic Analysis
PCAP	Packet Capture
RDP	Remote Desktop Protocol
SIEM	Security Information and Event Management
SOC	Security Operations Center
SSH	Secure Shell
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
VPN	Virtual Private Network

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Chapter 1

Introduction and Research Foundation

1.1 Background and Context

Modern organizations operate in highly connected digital environments where networks, cloud services, remote access systems, web applications, endpoints, and third-party integrations continuously expand the organizational attack surface. As a result, cyber threats have become increasingly complex, adaptive, and difficult to detect using traditional monitoring approaches alone. Modern cybersecurity risk management requires organizations to understand assets, risks, vulnerabilities, threats, and security outcomes in a structured manner [1].

Security Operations Centers (SOCs) play a central role in protecting organizational infrastructure by continuously monitoring security events, analyzing suspicious activities, detecting threats, and supporting timely incident response. Continuous monitoring is essential because organizations need ongoing awareness of information security, vulnerabilities, threats, and control effectiveness [2]. SOCs commonly rely on several security technologies, including Intrusion Detection Systems and Intrusion Prevention Systems (IDS/IPS), Security Information and Event Management (SIEM) platforms, firewalls, endpoint detection systems, vulnerability scanners, and threat intelligence sources.

Intrusion detection and prevention systems are important components of network security monitoring because they support the detection and prevention of suspicious or malicious activities across different deployment types [3]. In addition, effective log management supports security monitoring, incident investigation, and event correlation in SOC environments [4].

1.2 Research Motivation

The motivation for this research arises from the increasing gap between the complexity of modern cyberattacks and the ability of traditional SOC detection mechanisms to identify them quickly and accurately. Modern attackers often use multi-stage techniques that begin with reconnaissance and exploitation of exposed assets, followed by credential abuse, lateral move-

ment, command-and-control communication, or data exfiltration.

Reducing detection time is also important from an incident response perspective, because effective incident handling depends on timely detection, analysis, containment, eradication, and recovery activities [5, 6].

1.3 Research Problem

Despite significant advancements in cybersecurity monitoring technologies, many SOC environments still face limitations in detecting modern and complex attacks in a timely and accurate manner. Existing detection systems often operate using predefined signatures, static SIEM correlation rules, or isolated alerts generated by individual tools.

Therefore, the research problem is that current SOC detection approaches lack an integrated and time-aware methodology that combines real-time Attack Surface Management with network traffic analysis and leverages artificial intelligence to generate interpretable and dynamic detection rules capable of reducing attack detection time and improving alert accuracy.

Machine learning and anomaly-based approaches may support detection, but previous research has shown that intrusion detection systems still face challenges related to false positives, operational deployment, and unrealistic closed-world assumptions [7, 8, 9].

1.4 Problem Decomposition

To clarify the research problem, this study decomposes it into several connected components. Each component represents a limitation in current SOC detection practices and has a direct implication for the proposed hybrid methodology.

1.5 Research Gap

Existing research and operational practices have made important progress in intrusion detection, network traffic analysis, SIEM correlation, vulnerability management, and AI-based cybersecurity. However, these areas are frequently addressed as separate functions.

The research gap addressed by this study is the lack of a hybrid SOC-oriented methodology that combines real-time Attack Surface Management with network traffic analysis and uses artificial intelligence to generate dynamic detection rules that are both interpretable and operationally useful.

Risk assessment guidance emphasizes the importance of identifying threats, vulnerabilities, likelihood, impact, and risk as part of cybersecurity decision-making [10]. This supports the need to combine asset exposure and vulnerability context with observed network behavior.

Table 1.1: Problem Decomposition Summary

Problem Component	**Description**	**Research Implication**
Attack Surface Visibility Gap	SOC detection systems may not fully use information about exposed assets, open services, vulnerable systems, or attack paths.	The proposed methodology must include ASM data as a core input for detection rule generation.
Network Behavior Isolation	Network traffic analysis may detect suspicious patterns without understanding the exposure or criticality of the destination asset.	The methodology must correlate traffic behavior with asset exposure and risk context.
Static Detection Rules	Many IDS/SIEM rules are predefined and may not adapt quickly to changes in assets, services, or threats.	The study must support dynamic rule generation based on current attack surface and traffic evidence.
Slow Detection Time	Weak correlation between ASM and network telemetry may delay the identification of real attacks.	The methodology must aim to reduce Mean Time to Detect through contextual and early detection logic.
High False Positive Rate	Alerts generated without sufficient context may overwhelm SOC analysts.	The methodology must improve alert accuracy by using exposure-aware and behavior-aware rule conditions.
Limited AI Utilization	AI may be used for classification, but not always for explainable rule generation or SOC-oriented detection logic.	The study must define how AI supports risk scoring, prioritization, and interpretable detection rule generation.

1.6 Research Aim

This study aims to design and evaluate a hybrid methodology that integrates Attack Surface Management with network traffic analysis to generate AI-enhanced dynamic detection rules for reducing attack detection time and improving alert accuracy in SOC centers.

1.7 Research Objectives

To achieve the research aim, this study will pursue the following objectives:

1. To analyze the limitations of existing SOC detection approaches, with emphasis on static rules, delayed detection, false positives, and weak integration between ASM and network traffic analysis.
2. To identify the types of ASM data that are useful for contextual detection, including exposed assets, open ports, running services, vulnerabilities, asset criticality, and exposure status.
3. To identify suitable network traffic features that can represent suspicious behavior, such as connection frequency, source and destination patterns, port activity, protocol usage, flow duration, packet counts, byte counts, and time-window behavior.
4. To design a correlation mechanism that links ASM information with network traffic behavior to identify high-risk attack scenarios.
5. To develop an AI-enhanced risk scoring component that supports alert prioritization and dynamic detection rule generation.
6. To generate interpretable detection rules based on correlated ASM and traffic evidence.
7. To compare the proposed methodology with traditional static detection rules or baseline detection logic.
8. To evaluate the proposed methodology using metrics such as Mean Time to Detect, false positive rate, precision, recall, F1-score, and rule interpretability.

1.8 Research Questions

This study is guided by the following research questions:

1. What are the main limitations of existing SOC detection approaches in correlating attack surface information with real-time network traffic behavior?

2. Which ASM attributes are most useful for supporting contextual and dynamic detection rule generation?

3. Which network traffic features are suitable for identifying suspicious behavior related to exposed or vulnerable assets?

4. How can artificial intelligence support risk scoring and the generation of interpretable detection rules?

5. To what extent can the proposed hybrid methodology reduce attack detection time compared with traditional static detection rules?

6. To what extent can the proposed methodology reduce false positives and improve alert accuracy in SOC environments?

1.9 Research Hypotheses

Because this study follows a quantitative system evaluation design, the following hypotheses are proposed:

1. H1: Integrating ASM data with network traffic analysis improves attack detection context compared with using network traffic analysis alone.

2. H2: AI-enhanced risk scoring improves the prioritization of suspicious activities targeting exposed or vulnerable assets.

3. H3: Dynamic detection rules generated from correlated ASM and traffic evidence reduce Mean Time to Detect compared with traditional static detection rules.

4. H4: Context-aware detection rules reduce false positive alerts compared with generic detection rules that do not use attack surface context.

1.10 Expected Contribution and Novelty

The expected contribution of this study is a structured hybrid methodology for improving SOC detection through the integration of real-time Attack Surface Management, network traffic analysis, and AI-enhanced rule generation.

The novelty of the study lies in connecting three components that are often treated separately: ASM, network telemetry, and detection rule generation. Instead of using ASM only for vulnerability management, or using traffic analysis only for anomaly detection, the proposed methodology combines both sources to generate contextual and dynamic rules.

1.11 Significance of the Study

The significance of this study can be described from scientific, practical, and operational perspectives. Scientifically, the study contributes to cybersecurity research by proposing a hybrid methodology that connects attack surface visibility with behavioral network analysis and AI-supported rule generation.

Practically, the proposed methodology may help organizations improve detection quality by generating rules that are aligned with their actual attack surface and observed traffic behavior. Operationally, the study may support SOC analysts by reducing alert overload, improving prioritization, and shortening the time needed to identify high-risk activities.

1.12 Scope and Delimitations

The scope of this study is limited to the design and evaluation of a hybrid methodology for generating AI-enhanced detection rules using ASM data and network traffic analysis.

Table 1.2: Scope and Delimitations

Included in the Study	**Excluded from the Study**
Attack Surface Management data such as assets, ports, services, exposure, and vulnerabilities.	Full enterprise-scale ASM platform development.
Network traffic logs or flow-level records.	Payload decryption or deep packet inspection of encrypted content.
AI-based risk scoring and rule generation support.	Fully autonomous rule deployment without analyst review.
Dynamic and interpretable detection rule generation.	Offensive exploitation or unauthorized attack execution.
Evaluation using measurable SOC metrics.	Replacement of existing SIEM, IDS, or SOC platforms.
Controlled experimental or lab-based evaluation.	Full production SOC deployment.

1.13 Limitations

Several limitations are expected in this study. First, the research may rely on laboratory-generated data, public datasets, or controlled network traffic scenarios. These sources may not fully represent the complexity and noise of real enterprise SOC environments.

Second, the quality of ASM data may affect the effectiveness of the proposed methodology. Inaccurate asset inventories, outdated vulnerability information, or incomplete exposure data may reduce the quality of generated rules.

Third, AI-based scoring may require careful calibration to avoid overestimating or underestimating risk. Poorly calibrated risk scores may lead to ineffective prioritization or unnecessary alerts.

1.14 Operational Definition of Key Terms

Table 1.3: Operational Definition of Key Terms

Term	Operational Definition in This Study
SOC	A security operations environment responsible for monitoring, detecting, analyzing, and responding to cybersecurity threats.
Attack Surface Management	The process of identifying and monitoring assets, services, ports, vulnerabilities, and exposures that may be targeted by attackers.
Real-Time Attack Surface Mapping	A continuously updated representation of exposed assets, services, vulnerabilities, and risk conditions.
Network Traffic Analysis	The analysis of communication behavior using logs, flows, packets, or metadata such as IP addresses, ports, protocols, counts, and timing.
Network Telemetry	Security-relevant data collected from network devices, flows, firewalls, IDS/IPS, or monitoring sensors.
Dynamic Detection Rule	A detection rule generated or updated based on current ASM data and observed network behavior.
AI-Enhanced Detection	The use of artificial intelligence or machine learning to support risk scoring, classification, prioritization, or rule generation.
False Positive	An alert that identifies benign activity as malicious or suspicious.
Mean Time to Detect	The average time required to detect an attack or suspicious activity after it begins.
Rule Interpretability	The degree to which a generated detection rule can be understood, reviewed, and validated by a SOC analyst.

1.15 Organization of the Proposal

This proposal is organized into three main chapters. Chapter 1 presents the introduction and research foundation, including the background, motivation, research problem, problem decomposition, research gap, aim, objectives, research questions, hypotheses, expected contribution, significance, scope, limitations, and operational definitions.

Chapter 2 presents the literature review and research gap development. It reviews SOC operations, Attack Surface Management, network traffic analysis, IDS/IPS and SIEM detection rules, AI-based threat detection, dynamic rule generation, alert prioritization, false positive reduction, and Mean Time to Detect.

Chapter 3 describes the research methodology and experimental design. It explains the proposed framework architecture, ASM data collection, network traffic data collection, preprocessing, correlation logic, AI-based risk scoring, dynamic detection rule generation, baseline comparison, evaluation metrics, experimental environment, ethical considerations, threats to validity, expected outcomes, and proposed timeline.

Chapter 2

Literature Review and Research Gap Development

2.1 Introduction

This chapter reviews the main research areas related to the proposed study. The purpose of the literature review is to establish the theoretical and practical foundation for a hybrid methodology that combines Attack Surface Management (ASM), network traffic analysis, and artificial intelligence to generate dynamic detection rules for Security Operations Center (SOC) environments.

The chapter focuses on six main themes: SOC detection challenges, Attack Surface Management, network traffic analysis, IDS/IPS and SIEM rule-based detection, AI-enhanced cybersecurity detection, and dynamic detection rule generation. These themes are reviewed to identify the research gap addressed by the proposed study.

2.2 Security Operations Centers and Detection Challenges

Security Operations Centers are responsible for monitoring, detecting, analyzing, and responding to cybersecurity threats. Modern SOCs collect security-relevant events from multiple sources, including firewalls, IDS/IPS systems, endpoint agents, vulnerability scanners, SIEM platforms, and threat intelligence sources. Effective security monitoring requires continuous visibility into organizational assets, threats, vulnerabilities, and security control effectiveness [2].

Log management is also a core function in SOC environments. Security logs support detection, investigation, analysis, and response. However, organizations may face challenges related to log volume, inconsistent formats, storage, review, and correlation [4]. These challenges can make it difficult for SOC analysts to detect attacks quickly and accurately.

Although SOC tools provide important visibility, SOC environments still face several op-

erational challenges. First, analysts may receive a large number of alerts, many of which are false positives. Second, detection rules may be static and may not reflect the current exposure of organizational assets. Third, security tools may operate separately, which limits the ability to understand the full context of an attack.

2.3 Attack Surface Management

Attack Surface Management refers to the process of identifying and monitoring assets, services, ports, vulnerabilities, and exposures that may be targeted by attackers. In the context of cybersecurity risk management, organizations need to understand, assess, prioritize, and communicate cybersecurity risks [1]. ASM supports this need by providing visibility into exposed systems and potential entry points.

External Attack Surface Management is also recognized in practical cybersecurity guidance as a way to identify and manage internet-facing assets and exposures that may be targeted by attackers [11].

In this study, ASM data may include asset IP addresses, asset names, exposed ports, running services, service versions, vulnerability severity, exposure status, and asset criticality. This information provides useful context for detection because suspicious activity targeting an exposed or vulnerable asset may represent higher risk than the same activity targeting a non-critical or non-exposed asset.

However, ASM is often used mainly for vulnerability management or asset visibility. It is less commonly integrated directly with real-time network traffic analysis for dynamic detection rule generation. This creates an opportunity to use ASM as a contextual input for SOC detection.

2.4 Network Traffic Analysis and Network Telemetry

Network traffic analysis focuses on observing communication behavior between systems. It may use packet captures, flow records, firewall logs, IDS alerts, NetFlow-like metadata, or SIEM events. Useful traffic features may include source IP, destination IP, destination port, protocol, connection count, byte count, packet count, flow duration, time window, and connection frequency.

Network telemetry is important because it shows what is actually happening in the network. For example, repeated connection attempts to SSH, RDP, VPN, or web services may indicate reconnaissance, brute-force attempts, exploitation attempts, or abnormal access behavior.

However, network traffic analysis alone may not provide enough context. A traffic pattern may look suspicious, but its true risk depends on the exposure, vulnerability, and criticality of the destination asset. Therefore, combining traffic behavior with ASM information can support

more accurate and contextual detection.

2.5 IDS/IPS and SIEM Detection Rules

IDS/IPS systems and SIEM platforms commonly use rules to detect suspicious or malicious behavior. These rules may be based on signatures, thresholds, event correlation, indicators of compromise, or known attack patterns. Rule-based detection is useful because rules are interpretable and can be reviewed by SOC analysts.

MITRE ATT&CK provides a structured knowledge base of adversary tactics and techniques that can support threat-informed detection engineering and mapping of attack behaviors [12]. Such knowledge bases can help SOC teams understand attacker behavior and design detection logic aligned with known techniques.

Despite their usefulness, traditional detection rules have limitations. They are often manually created, static, and generic. A generic rule may not consider whether the targeted asset is exposed, vulnerable, or critical. This may increase false positives or delay detection of high-risk activities.

The proposed study addresses this limitation by using ASM and network traffic analysis together to generate more contextual and dynamic detection rules.

2.6 Artificial Intelligence in Cybersecurity Detection

Artificial intelligence and machine learning have been widely studied in cybersecurity for classification, anomaly detection, clustering, risk scoring, and alert prioritization. Buczak and Guven [8] reviewed data mining and machine learning methods for cyber security intrusion detection, showing the importance of data-driven approaches in cyber analytics.

Anomaly detection is also relevant to SOC environments because attacks may appear as deviations from normal behavior. Chandola, Banerjee, and Kumar [9] provide a broad survey of anomaly detection techniques, which supports the use of behavioral analysis for suspicious activity detection.

However, the use of machine learning in network intrusion detection must be treated carefully. Sommer and Paxson [7] highlighted challenges related to applying machine learning to intrusion detection, including unrealistic assumptions and difficulties in operational deployment. Therefore, this study does not treat AI as a replacement for SOC analysts. Instead, AI is used as a supporting component for risk scoring and interpretable detection rule generation.

2.7 Dynamic Detection Rule Generation

Dynamic detection rule generation refers to creating or updating detection rules based on current security context rather than relying only on static predefined rules. In the proposed study, rules are generated based on the relationship between ASM data and observed network traffic behavior.

The process can be summarized as follows:

1. Collect ASM data about assets, services, ports, exposure, and vulnerabilities.
2. Collect network traffic data or network telemetry.
3. Correlate traffic behavior with ASM context.
4. Calculate a risk score using AI or rule-assisted logic.
5. Generate an interpretable detection rule.
6. Evaluate the generated rule using SOC-relevant metrics.

This approach may improve detection because the generated rules are linked to the actual attack surface and current traffic behavior of the organization.

2.8 Summary of Related Research Areas

Table 2.1 summarizes the main research areas related to this study.

2.9 Research Gap Development

The reviewed areas show that SOCs already use several important security technologies, including SIEM platforms, IDS/IPS systems, ASM tools, and network monitoring solutions. However, these technologies are often used separately. ASM provides knowledge about exposed and vulnerable assets, while network traffic analysis provides knowledge about real-time communication behavior. When these two sources are not integrated, the detection system may miss important context.

The key research gap is the lack of a hybrid SOC-oriented methodology that combines real-time ASM data with network traffic analysis and uses AI to generate interpretable and dynamic detection rules.

This gap is important because attack detection should not depend only on generic signatures or static thresholds. Detection rules should reflect both the risk of the targeted asset and the behavior observed in the network.

Table 2.2 presents the research gap development matrix.

Table 2.1: Summary of Related Research Areas

Research Area	**Main Focus**	**Relevance to This Study**
SOC Detection	Monitoring alerts, logs, incidents, and security events.	Provides the operational environment where the proposed methodology will be applied.
Attack Surface Management	Identifying exposed assets, open ports, services, vulnerabilities, and risks.	Provides contextual information for prioritizing suspicious traffic and generating rules.
Network Traffic Analysis	Analyzing communication behavior using flows, logs, packets, or metadata.	Provides real-time behavioral evidence about possible attacks.
IDS/IPS and SIEM Rules	Detecting suspicious activities using signatures, thresholds, and correlation rules.	Provides the rule-based detection output targeted by the proposed methodology.
Artificial Intelligence	Supporting classification, anomaly detection, risk scoring, and prioritization.	Supports dynamic and context-aware rule generation.
Dynamic Rule Generation	Creating or updating detection rules based on changing data and risk context.	Represents the main output of the proposed methodology.

Table 2.2: Research Gap Development Matrix

Current Limitation	**Impact on SOC Detection**	**Proposed Direction**
ASM and traffic data are often analyzed separately.	Limited attack context and weaker alert prioritization.	Correlate ASM data with network traffic behavior.
Detection rules are often static.	Rules may not adapt to changes in assets, services, and threats.	Generate dynamic rules based on current exposure and traffic patterns.
Generic rules may produce false positives.	Analysts may experience alert fatigue and delayed response.	Use asset exposure and traffic context to improve rule specificity.
AI is often used for classification only.	Limited support for explainable SOC-oriented rule generation.	Use AI for risk scoring and interpretable rule generation support.
Detection time may be delayed.	Attacks may progress before being detected.	Evaluate the methodology using Mean Time to Detect and alert accuracy metrics.

2.10 Conceptual Framework of the Proposed Study

The conceptual framework of this study is based on the integration of three main inputs and one main output. The first input is ASM data, which describes exposed assets, ports, services, vulnerabilities, and asset criticality. The second input is network traffic data, which describes communication behavior. The third component is AI-enhanced analysis, which supports risk scoring and prioritization.

The main output is a set of interpretable detection rules that can be reviewed and used by SOC analysts. The relationship can be described as follows:

ASM Data + Network Traffic Analysis + AI-Based Risk Scoring = Dynamic Detection Rules

This framework supports the main assumption of the study: detection rules become more useful when they are generated from both attack surface context and real-time traffic behavior.

2.11 Chapter Summary

This chapter reviewed the main concepts related to the proposed study, including SOC detection challenges, Attack Surface Management, network traffic analysis, IDS/IPS and SIEM rules, AI-based detection, and dynamic detection rule generation. The chapter identified a research gap in the limited integration between ASM and network telemetry for generating contextual detection rules.

The next chapter presents the research methodology and experimental design. It explains how ASM data and network traffic data will be collected, correlated, processed, and used to generate AI-enhanced detection rules. It also defines the evaluation metrics that will be used to assess detection time reduction and alert accuracy.

Chapter 3

Research Methodology and Experimental Design

3.1 Introduction

This chapter presents the research methodology and experimental design of the proposed study. The study aims to design and evaluate a hybrid methodology that integrates real-time Attack Surface Management (ASM) data with network traffic analysis to generate AI-enhanced dynamic detection rules for SOC environments.

The chapter explains the research design, proposed framework, data sources, preprocessing steps, correlation logic, AI-enhanced risk scoring, detection rule generation, baseline comparison, evaluation metrics, experimental environment, ethical considerations, threats to validity, and expected outcomes.

3.2 Research Design

This study follows a quantitative computing research design based on algorithm and system evaluation. The focus is not on surveys or interviews, but on designing a technical framework and evaluating its performance using measurable indicators.

This design is aligned with security monitoring and log analysis practices, where measurable evidence is collected, processed, correlated, and evaluated to support detection and response activities [2, 4].

The proposed methodology will be evaluated by comparing two detection approaches:

1. A traditional detection approach based on static or generic detection rules.

2. The proposed hybrid approach that combines ASM data, network traffic analysis, and AI-enhanced risk scoring to generate dynamic detection rules.

The comparison will focus on whether the proposed approach can reduce attack detection time, improve alert accuracy, and reduce false positives.

3.3 Proposed Framework Overview

The proposed framework consists of five main stages:

1. ASM data collection.
2. Network traffic data collection.
3. Data preprocessing and feature extraction.
4. Correlation between ASM data and network traffic behavior.
5. AI-enhanced risk scoring and dynamic detection rule generation.

The general idea of the framework is to detect whether suspicious network behavior is targeting exposed, vulnerable, or critical assets. If a high-risk relationship is identified, the system generates an interpretable detection rule that can be reviewed and used by SOC analysts.

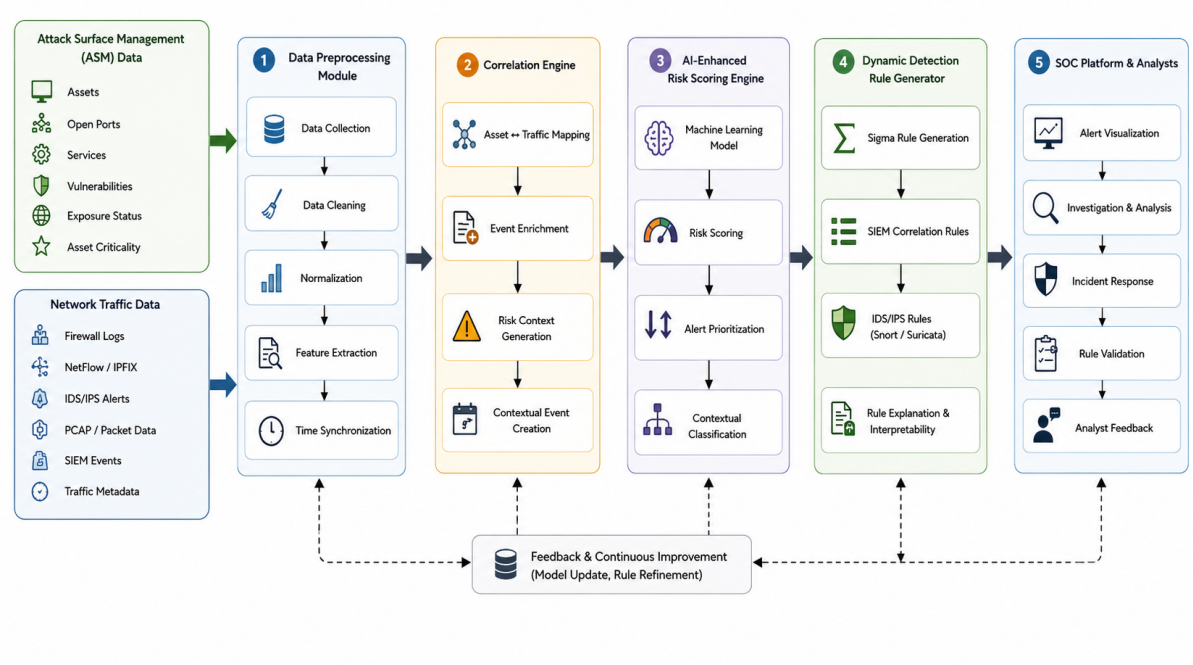


Figure 3.1: Proposed Hybrid SOC Detection Architecture

Figure 3.1 illustrates the main components of the proposed framework and the interaction between ASM data, network traffic analysis, risk scoring, detection rule generation, and SOC analyst feedback.

3.4 Data Sources

The proposed study will use two main categories of data: ASM data and network traffic data.

3.4.1 Attack Surface Management Data

ASM data represents information about assets and services that may be targeted by attackers. In this study, ASM data may include:

- Asset IP address.
- Asset name.
- Open ports.
- Running services.
- Protocol type.
- Exposure status.
- Vulnerability severity.
- Asset criticality.

This data can be collected from asset inventories, vulnerability scanning tools, port scanning tools, or manually prepared CSV files for experimental purposes. The ASM data will be used to provide security context for the traffic analysis stage.

Information security testing and assessment guidance supports the use of technical assessment methods such as vulnerability scanning and network discovery to identify weaknesses and validate security posture [13]. Therefore, ASM data in this study can be prepared from authorized scanning outputs, asset inventories, or controlled experimental templates.

3.4.2 Network Traffic Data

Network traffic data represents communication behavior observed in the network. This data may be collected from firewall logs, IDS alerts, NetFlow-like records, packet captures, or SIEM events. In this study, the traffic data may include:

- Source IP address.
- Destination IP address.
- Destination port.
- Protocol.

- Number of connections.
- Flow duration.
- Packet count.
- Byte count.
- Time window.
- Traffic direction.

The network traffic data will be analyzed to identify suspicious patterns such as repeated connection attempts, abnormal port access, high connection frequency, or unusual communication behavior.

3.5 Data Preprocessing

Before analysis, both ASM data and network traffic data will be cleaned and prepared. The preprocessing stage includes:

1. Removing incomplete or duplicated records.
2. Normalizing IP address and port formats.
3. Converting timestamps into a standard format.
4. Aggregating traffic records by time window.
5. Mapping destination IP addresses and ports to ASM assets.
6. Encoding categorical values such as service type and exposure level.

This stage ensures that ASM data and network traffic data can be correlated correctly.

3.6 Feature Selection

The proposed methodology will use features from both ASM and network traffic data. Table 3.1 summarizes the feature categories used in the study.

Table 3.1: Feature Categories Used in the Proposed Methodology

Feature Category	Example Features	Purpose
ASM Features	Asset criticality, exposed port, service type, vulnerability severity, exposure status.	To provide context about the risk level of the targeted asset.
Traffic Features	Source IP, destination IP, destination port, protocol, connection count, packet count, byte count, time window.	To represent observed network behavior.
Correlation Features	Match between traffic destination and exposed asset, access to high-risk service, repeated attempts to vulnerable service.	To identify high-risk relationships between exposure and behavior.
Risk Features	Combined risk score, anomaly level, priority level.	To support alert prioritization and rule generation.

3.7 Correlation Logic

The core of the proposed methodology is the correlation between ASM data and network traffic behavior. The system checks whether suspicious traffic is targeting an exposed, vulnerable, or critical asset.

For example, if ASM data shows that a server has RDP exposed on port 3389 and the traffic data shows repeated connection attempts to that port from external sources, the system treats this as a high-risk scenario.

The correlation logic can be described as follows:

Exposed Asset + Suspicious Traffic Behavior = High-Risk Detection Scenario

The correlation stage will produce enriched records that combine both asset context and traffic behavior.

3.8 AI-Enhanced Risk Scoring

The AI-enhanced risk scoring component will assign a risk score to each correlated event or scenario. The score may be based on features such as asset criticality, exposure status, vulnerability severity, number of connection attempts, service type, traffic direction, and time-window behavior. Machine learning has been widely studied for intrusion detection and cybersecurity analytics [8]; however, its operational use must be carefully designed because intrusion detection environments are complex, dynamic, and affected by false positives and changing attack behavior [7].

The output of this stage may classify each scenario into one of three levels:

- Low risk.
- Medium risk.
- High risk.

The AI component may use a machine learning classifier or a rule-assisted scoring model. The purpose of AI in this study is not to replace the SOC analyst, but to support prioritization and rule generation.

3.9 Dynamic Detection Rule Generation

After risk scoring, the system generates interpretable detection rules for high-risk scenarios. These rules may be written in a generic rule format, SIEM correlation logic, or IDS-like rule format.

For example, if the system detects repeated attempts against an exposed SSH service, it may generate a rule such as:

> Alert when the same source IP attempts to connect to an exposed SSH service more than 20 times within 60 seconds.

The generated rules should be understandable and reviewable by SOC analysts. This is important because SOC environments require explainable detection logic rather than black-box outputs only.

3.10 Baseline Comparison

To evaluate the proposed methodology, it will be compared with a baseline approach. The baseline approach may use static detection rules that do not include ASM context.

The baseline comparison is necessary because traditional IDS/IPS and SIEM-based detection approaches may rely on predefined rules or signatures, while the proposed methodology adds asset exposure and risk context to the detection process [3].

The comparison will help answer whether integrating ASM data with network traffic analysis improves detection performance. Table 3.2 summarizes the comparison plan.

3.11 Experimental Protocol

The experimental protocol will include the workflow illustrated in Figure 3.2.

Table 3.2: Baseline Comparison Plan

Approach	Description	Expected Limitation
Baseline Detection	Uses static or generic rules based mainly on network traffic thresholds.	May generate false positives and may not consider asset exposure or vulnerability context.
Proposed Hybrid Detection	Uses ASM data, network traffic analysis, correlation logic, and AI-enhanced risk scoring.	Requires accurate ASM data and careful risk score calibration.

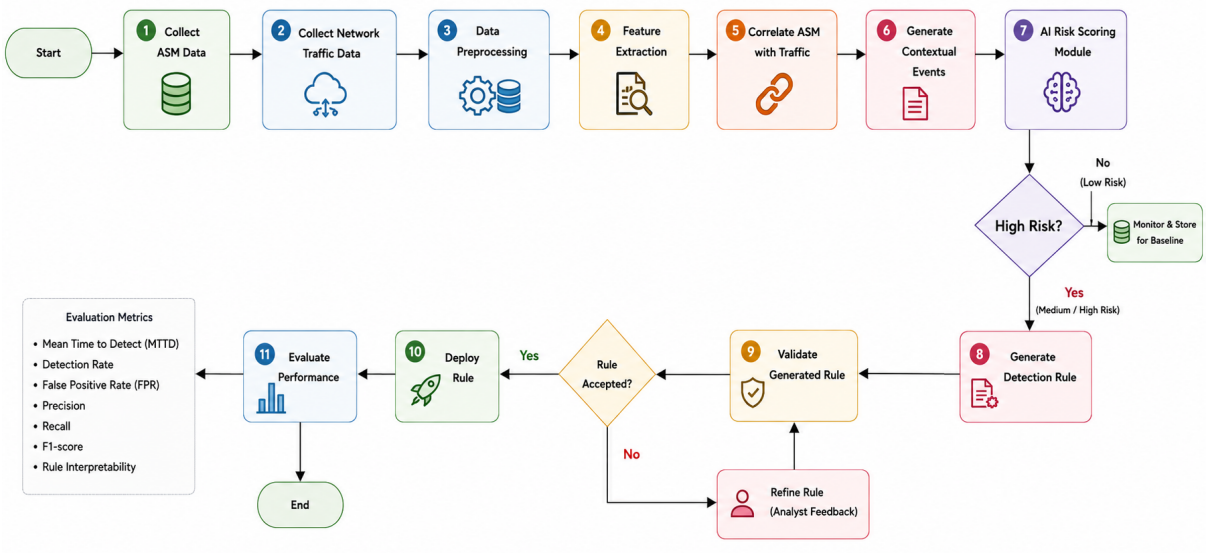


Figure 3.2: Proposed Methodology Flowchart

Figure 3.2 summarizes the complete workflow of the proposed methodology, starting from data collection and preprocessing, followed by feature extraction, ASM–traffic correlation, AI-enhanced risk scoring, dynamic rule generation, rule validation, deployment, and performance evaluation.

The experimental procedure consists of the following steps:

1. Prepare the ASM dataset containing assets, services, ports, vulnerabilities, and risk levels.
2. Prepare the network traffic dataset containing communication records and suspicious behavior scenarios.
3. Apply preprocessing to clean and normalize both datasets.
4. Extract relevant features from both ASM and traffic data.
5. Correlate ASM information with observed network traffic.
6. Apply AI-enhanced risk scoring to prioritize correlated events.
7. Generate dynamic detection rules for high-risk scenarios.
8. Validate and deploy the generated rules.
9. Compare the proposed approach with the baseline detection approach.
10. Evaluate the methodology using Mean Time to Detect (MTTD), Detection Rate, False Positive Rate, Precision, Recall, F1-score, and Rule Interpretability.

This protocol provides a systematic and reproducible procedure for evaluating the effectiveness of the proposed hybrid methodology against conventional rule-based detection approaches.

3.12 Evaluation Metrics

The proposed methodology will be evaluated using SOC-relevant metrics. Table 3.3 summarizes the main metrics.

The use of multiple metrics is important because accuracy alone may not provide sufficient evidence of detection quality, especially when false positives and missed attacks are operationally significant [7, 9].

Reducing detection time is important because incident response guidance emphasizes the need to improve the efficiency and effectiveness of detection, response, and recovery activities [6].

Table 3.3: Evaluation Metrics

Metric	Purpose
Mean Time to Detect	Measures the average time required to detect an attack or suspicious activity.
Detection Rate	Measures the percentage of actual attacks or suspicious scenarios detected by the system.
False Positive Rate	Measures the percentage of benign activities incorrectly classified as suspicious.
Precision	Measures how many generated alerts are actually relevant or correct.
Recall	Measures how many actual attack scenarios were successfully detected.
F1-score	Provides a balanced measure between precision and recall.
Rule Interpretability	Assesses whether the generated rule can be understood and reviewed by SOC analysts.

3.13 Experimental Environment

The experimental environment may include a laboratory network, public datasets, or simulated CSV-based datasets. The study may use tools such as Python for data processing, machine learning, risk scoring, and rule generation.

The environment will document the following:

- Operating system.
- Programming language and libraries.
- Dataset sources.
- Preprocessing steps.
- Model or scoring settings.
- Evaluation scripts.

Documenting the experimental environment is important for reproducibility and fair evaluation.

3.14 Ethical Considerations

This study will avoid unauthorized scanning, exploitation, or collection of private network data. Any experimental data used in the study should be collected from authorized laboratory environments, public datasets, or simulated data prepared for research purposes.

The study does not aim to perform offensive exploitation. Its purpose is to improve defensive SOC detection by generating contextual and interpretable detection rules.

The defensive focus of this study is consistent with incident response and cybersecurity risk management practices that aim to improve organizational preparation, detection, response, and recovery capabilities [6].

3.15 Threats to Validity

Several threats to validity may affect the study. First, laboratory or simulated data may not fully represent real SOC environments. Second, inaccurate ASM data may reduce the quality of correlation and rule generation. Third, AI-based risk scoring may require careful calibration to avoid misleading results. Fourth, results may depend on the selected attack scenarios, thresholds, and datasets.

These threats will be addressed by documenting assumptions, using clear evaluation metrics, comparing with a baseline approach, and explaining the limitations of the proposed methodology.

A further validity threat is the accuracy of asset and vulnerability information. If ASM data is incomplete or outdated, the correlation process may produce inaccurate risk scores or detection rules. This threat will be mitigated by documenting the data source, preprocessing steps, and assumptions used during experimentation [10, 13].

3.16 Experimental Variables

This study includes independent, dependent, and control variables. Defining these variables is important to support a clear and measurable evaluation of the proposed methodology.

3.17 Proposed Timeline

The proposed timeline organizes the research activities into sequential phases. The timeline may be adjusted depending on dataset availability, implementation progress, and supervisor feedback.

3.18 Expected Outcomes

The expected outcomes of this study are:

1. A hybrid methodology that integrates ASM data with network traffic analysis.
2. A correlation mechanism that links exposed assets with suspicious traffic behavior.

Table 3.4: Experimental Variables

Variable Type	**Variable**	**Description**
Independent Variable	Detection Method	The detection approach used in the experiment, either traditional static detection rules or the proposed hybrid ASM-based methodology.
Independent Variable	ASM Context Usage	Whether asset exposure, service risk, vulnerability severity, and asset criticality are included in the detection process.
Dependent Variable	Mean Time to Detect	The time required to detect suspicious or malicious behavior after it appears in the traffic data.
Dependent Variable	False Positive Rate	The percentage of benign activities incorrectly classified as suspicious.
Dependent Variable	Precision, Recall, and F1-score	Quantitative metrics used to evaluate detection accuracy and alert quality.
Control Variable	Dataset or Scenario	The same traffic dataset or attack scenario will be used when comparing the baseline and proposed approach.
Control Variable	Time Window	The same aggregation window will be applied when comparing detection results.
Control Variable	Threshold Policy	Thresholds will be documented and applied consistently during evaluation.

Table 3.5: Proposed Research Timeline

Period	Planned Activity
Month 1	Finalize research proposal, refine research questions, and confirm scope with the supervisor.
Month 2	Conduct detailed literature review on SOC detection, ASM, network traffic analysis, AI-based detection, and dynamic rule generation.
Month 3	Prepare ASM data template, network traffic data template, and experimental dataset structure.
Month 4	Implement preprocessing steps and ASM-to-traffic correlation logic.
Month 5	Develop AI-enhanced risk scoring and dynamic detection rule generation components.
Month 6	Define baseline detection rules and run initial comparison experiments.
Month 7	Evaluate the proposed methodology using MTTD, false positive rate, precision, recall, F1-score, and detection rate.
Month 8	Analyze results, document findings, discuss limitations, and refine the framework.
Month 9	Write final thesis chapters and prepare tables, appendices, and references.
Month 10	Final review, formatting, supervisor feedback, corrections, and submission preparation.

3. An AI-enhanced risk scoring component.

4. A dynamic detection rule generation process.

5. An evaluation of detection time, false positives, and alert accuracy.

3.19 Chapter Summary

This chapter presented the research methodology and experimental design of the proposed study. It described the research design, proposed framework, data sources, preprocessing, feature selection, correlation logic, AI-enhanced risk scoring, dynamic rule generation, baseline comparison, experimental protocol, evaluation metrics, ethical considerations, threats to validity, and expected outcomes.

The next section of the proposal presents the conclusion and prepares the study for implementation and evaluation.

The study also follows a defensive and privacy-aware research direction by avoiding unauthorized exploitation, unauthorized scanning, or collection of private traffic outside approved environments.

Conclusion

This research proposal presented a hybrid methodology for integrating real-time Attack Surface Management (ASM) data with network traffic analysis to generate AI-enhanced dynamic detection rules for Security Operations Center (SOC) environments. The proposed study is motivated by the limitations of current SOC detection approaches, particularly delayed attack detection, high false positive rates, static detection rules, and weak correlation between asset exposure and real-time network behavior.

The proposal identified a research problem in the absence of an integrated and time-aware methodology that combines attack surface visibility with network telemetry to support contextual and interpretable detection rule generation. To address this problem, the study proposed a framework that uses ASM data, network traffic features, correlation logic, AI-enhanced risk scoring, and dynamic detection rule generation.

The proposed methodology is designed to support SOC analysts rather than replace them. Its main purpose is to improve detection context by linking suspicious traffic behavior with exposed, vulnerable, or critical assets. This can help analysts prioritize alerts, reduce unnecessary false positives, and detect attacks earlier than traditional static rule-based approaches.

The study follows a quantitative computing research design based on algorithm and system evaluation. The proposed methodology will be compared with a baseline detection approach that uses static or generic rules. The evaluation will use SOC-relevant metrics such as Mean Time to Detect, false positive rate, precision, recall, F1-score, detection rate, and rule interpretability.

Overall, the expected contribution of this study is a structured, practical, and SOC-oriented methodology for generating dynamic detection rules from integrated ASM and network traffic evidence. The research is expected to support more contextual, adaptive, and explainable detection engineering in modern SOC environments.

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Appendix A

ASM Data Template

This appendix presents the proposed structure for Attack Surface Management (ASM) data used in the study. The purpose of this template is to standardize asset exposure information before correlating it with network traffic behavior.

Field	**Description**	**Example**
Asset ID	Unique identifier for the asset.	ASSET-001
Asset IP	IP address of the monitored asset.	192.168.1.10
Asset Name	Descriptive name of the asset.	Web Server
Service	Running service on the asset.	SSH, RDP, HTTP
Port	Open port associated with the service.	22, 3389, 80
Protocol	Transport protocol used by the service.	TCP or UDP
Exposure Status	Whether the service is exposed internally or externally.	Internal, External
Vulnerability Severity	Severity level of known vulnerability or weakness.	Low, Medium, High, Critical
Asset Criticality	Business or operational importance of the asset.	Low, Medium, High
Risk Level	Combined risk level based on exposure and criticality.	Medium, High, Critical

Appendix B

Network Traffic Data Template

This appendix presents the proposed structure for network traffic data used in the study. The purpose of this template is to standardize network behavior records before correlation with ASM data.

Field	Description	Example
Timestamp	Time when the traffic record was observed.	2026-01-21 10:30:00
Source IP	IP address initiating the connection.	185.10.10.5
Destination IP	IP address receiving the connection.	192.168.1.10
Destination Port	Port targeted by the connection.	22
Protocol	Network protocol used.	TCP
Connection Count	Number of connections within a time window.	50
Packet Count	Number of packets observed.	300
Byte Count	Number of bytes transferred.	25000
Time Window	Aggregation period for analysis.	60 seconds
Traffic Direction	Direction of communication.	External-to-Internal
Observed Behavior	Description of the traffic pattern.	Repeated SSH attempts

Appendix C

Dynamic Detection Rule Template

This appendix presents a generic template for detection rules generated by the proposed methodology. The generated rules should be interpretable and reviewable by SOC analysts before operational use.

Rule Element	Description
Rule ID	Unique identifier for the generated rule.
Rule Name	Short descriptive name of the rule.
Description	Explanation of the suspicious behavior detected by the rule.
Target Asset	The asset affected by the rule.
Target Service	The exposed or monitored service.
Detection Condition	Logical condition used to trigger the alert.
Threshold	Number or value required to trigger the rule.
Time Window	Time period used to evaluate the condition.
Risk Score	AI-enhanced or rule-assisted risk score assigned to the scenario.
Severity	Alert severity level such as Low, Medium, High, or Critical.
Recommended Action	Suggested action for SOC analysts.

Example Rule

Rule Name: Possible SSH Brute Force Against Exposed Asset

Condition: Alert when the same source IP attempts to connect to an exposed SSH service more than 20 times within 60 seconds.

Severity: High

Recommended Action: Review source IP reputation, validate authentication logs, check whether the destination asset is externally exposed, and block or investigate

the source if malicious behavior is confirmed.

Appendix D

Evaluation Checklist

This appendix provides a checklist for evaluating the proposed methodology during implementation and testing.

No.	Evaluation Item	Status / Notes
1	ASM dataset includes asset IP, service, port, exposure status, vulnerability severity, and asset criticality.	
2	Network traffic dataset includes source IP, destination IP, destination port, protocol, connection count, and time window.	
3	Data preprocessing steps are documented.	
4	ASM records are correctly mapped to traffic records.	
5	Correlation logic is clearly defined.	
6	AI-enhanced risk scoring method is explained.	
7	Generated detection rules are interpretable.	
8	Baseline detection approach is defined.	
9	Evaluation metrics are calculated.	
10	Threats to validity and limitations are documented.	

Appendix E

Risk Register Template

This appendix presents a risk register template for documenting possible risks during the implementation and evaluation of the proposed study.

Risk	Description	Impact	Mitigation
Incomplete ASM Data	Asset or exposure information may be missing or outdated.	High	Validate ASM data before correlation and document assumptions.
Noisy Traffic Data	Network logs may contain incomplete, duplicated, or irrelevant records.	Medium	Apply preprocessing and filtering before analysis.
False Positives	Generated rules may alert on benign behavior.	High	Use ASM context, thresholds, and analyst review to improve rule quality.
Poor Risk Score Calibration	AI or scoring model may overestimate or underestimate risk.	Medium	Test multiple thresholds and compare results with baseline rules.
Limited Dataset Realism	Lab or simulated data may not fully represent real SOC environments.	Medium	Clearly document dataset limitations and experimental assumptions.
Rule Interpretability Issue	Generated rules may be difficult for analysts to understand.	Medium	Use readable rule templates and include explanation fields.